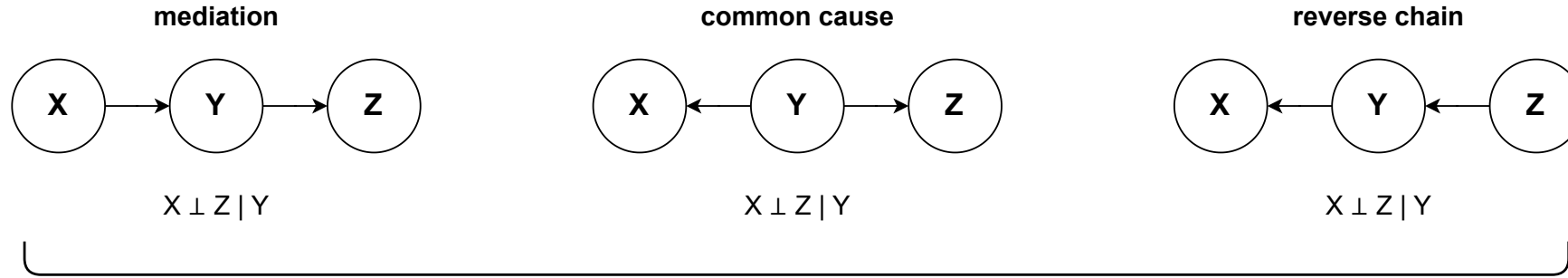


Markov-Equivalence Problem

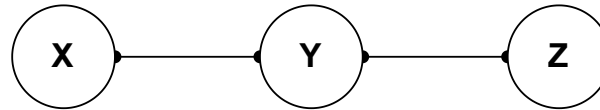
A cross-sectional table leaves (algorithms) no way to orient these edges.



same conditional-independence pattern



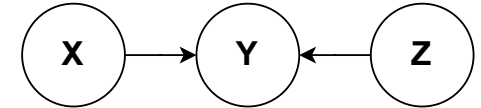
CPDAG RETURNED



Everything inside the equivalence class is observationally indistinguishable.

Recovering orientation beyond the CPDAG requires signal the flat table discards: **temporal order, repeated events, or interventions.**

Collider



$X \perp Z$ (marginal independence)

$X \not\perp Z | Y$ (**conditioning on Y opens the path**)